Facilities, Equipment, and Other Resources

1. Resources at the University of California, Santa Barbara

The University of California, Santa Barbara (UCSB) is a leading research institution classified as a Research 1 University and is a member of the Association of American Universities. UCSB College of Engineering and the Computer Science Department are highly ranked. UCSB College of Engineering and the Department of Computer Science (CS) provide ample resources to PIs and graduate researchers to conduct research.

UCSB Computer Science Department facilities are based on a client/server architecture. The Department maintains a large network of computers and servers running Linux, Mac OS, and Windows. The Computer Science Instructional Laboratory (CSIL) serves all Computer Science students' course and lab needs, while the Graduate Student Laboratory (GSL) is reserved exclusively for graduate students. In addition to these student labs, there are more than twenty research labs throughout the Department. The CS department has a high-performance computing server with 4 nodes and 8 NVIDIA A100 GPUs on each node.

In addition to the office and laboratory spaces, the PI and students have access to a state-of-the-art graphics processing unit (GPU) computer cluster at the Center for Scientific Computing (CSC) at UCSB, and to the UCSB campus cloud that is part of the NSF Aristotle project. The CSC at the California NanoSystems Institute provides a number of HPC systems as well as training using the best computing resources. The staff all have a background in science or math, and a number of years of experience in assisting and teaching users and providing expertise in porting, developing, and optimizing code. The following clusters are available to any researcher on campus: a) an HPC system with 126 nodes (1,500 cores) of Infiniband interconnected MPI nodes, four "fat nodes" of 512/1024GB of RAM, and 12 M2050GPU systems; 150 TB (usable) high performance storage; b) an HPC system (2,600 cores), acquired in 2018, with 64 regular computing nodes, each with 40 cores and 192GB of RAM, four "fat nodes" of 1TB of RAM, three GPU nodes with 4xNVIDIA V100/32GB GPUs/node with NVLINK, and a 600TB parallel filesystem.

Lab space. PI Vigna and PI Kruegel are the directors of the Security Laboratory (SecLab) located on the 2nd floor of the Harold Frank Hall at UCSB. SecLab provides desk and meeting spaces for researchers and is able to host up to 20 researchers.

Computers & HPC. SecLab has a machine room with space for all requested server clusters and their necessary infrastructure. SecLab also has access to several large-scale Linux workstations and servers.

2. Available Resources at Arizona State University

As an Association of American Universities (AAU) member and Carnegie R1 institution exceeding $800 million in annual research expenditures, Arizona State University offers an elite environment for scientific discovery.

The Center for Cybersecurity and Trusted Foundations (CTF), as part of ASU's Advanced Capabilities for National Security Institute, is a multi-disciplinary center that brings together leading faculty to conduct sponsored research focused on binary analysis, identity management, privacy issues, malware attribution, secure mobile devices, predictive analytics/adversarial dynamics, and vulnerability analysis; to create custom education, internship and training platforms; and to support entrepreneurial activities in partnership with ASU SkySong.

Current research activities of the center involve malware reverse-engineering, next-generation authentication, identity management, privacy and ethics, the human connection, dark web economics, hacking competitions, automated binary analysis systems, humans collaborating with automated cyber reasoning systems, and advanced decompilation. CTF research projects are driven by real-world challenges and result in open-source prototypes.

Dr. Doupé is the Center's Director and is a co-director, alongside Dr. Shoshitaishvili, of the Laboratory of Security Engineering for Future Computing (SEFCOM: http://sefcom.asu.edu). SEFCOM occupies a designated research area in the center to conduct experimental research. Current equipment consists of 1 DNS server, 1 DHCP server, 1 ARM DSTREAM High-Performance Debug and Trace unit, 40 high-end servers, and 30 PC systems with Windows and other UNIX systems including nomadic-computing related equipment, which includes RF-ID tags, a GPS system, wireless sensors, beacons, tablet PCs, Android handhelds, Google TVs, and Tablet PCs.

The researchers will utilize not only the hardware resources of the CTF and SEFCOM to perform the proposed project, but also the interdisciplinary connections that CTF provides, to create cross-discipline collaborations that further the goals of the proposal.